Trabajo en clase # 08 (llaves OpenVPN)

Certificado auto-firmado

1. Instalar easy-rsa

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
jeferson@jeferson-IdeaPad-3-15ITL6:~$ sudo apt  install easy-rsa
[sudo] password for jeferson:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  zorin-os-feedback
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libccid libeac3 opensc opensc-pkcs11 pcscd
The following NEW packages will be installed:
  easy-rsa libccid libeac3 opensc opensc-pkcs11 pcscd
0 upgraded, 6 newly installed, 0 to remove and 11 not upgraded.
Need to get 1 570 kB of archives.
After this operation, 5 046 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://cr.archive.ubuntu.com/ubuntu noble/universe amd64 libccid amd64 1.5
.5-1 [83,0 kB]
Get:2 http://cr.archive.ubuntu.com/ubuntu noble/universe amd64 pcscd amd64 2.0.3
-1build1 [61,7 kB]
Get:3 http://cr.archive.ubuntu.com/ubuntu noble/universe amd64 libeac3 amd64 1.1
.2+ds+git20220117+453c3d6b03a0-1.1build2 [54,3 kB]
Get:4 http://cr.archive.ubuntu.com/ubuntu noble-updates/universe amd64 opensc-pk
cs11 amd64 0.25.0~rc1-1ubuntu0.2 [936 kB]
```

2. Iniciar PKI

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
/usr/share/easy-rsa/easyrsa
jeferson@jeferson-IdeaPad-3-15ITL6:~$ /usr/share/easy-rsa/easyrsa init-pki

WARNING!!!

You are about to remove the EASYRSA_PKI at:
* /home/jeferson/pki

and initialize a fresh PKI here.

Type the word 'yes' to continue, or any other input to abort.
  Confirm removal: yes


Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /home/jeferson/pki

Using Easy-RSA configuration:
* undefined

jeferson@jeferson-IdeaPad-3-15ITL6:~$
```

3. Crear la autoridad certificadora [CA]

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:CI0123 CA

Notice
------
CA creation complete. Your new CA certificate is at:
* /home/jeferson/pki/ca.crt

jeferson@jeferson-IdeaPad-3-15ITL6:~$
```

4. Generar la llave de encripción [DH]

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
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```

5. Generar la llave para el servidor

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
Type the word 'yes' to continue, or any other input to abort.
  Confirm request details: yes

Using configuration from /home/jeferson/pki/openssl-easyrsa.cnf
Enter pass phrase for /home/jeferson/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'redes'
Certificate is to be certified until Jul 26 17:34:31 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
------
Certificate created at:
* /home/jeferson/pki/issued/redes.crt

Notice
------
Inline file created:
* /home/jeferson/pki/inline/redes.inline

jeferson@jeferson-IdeaPad-3-15ITL6:~$
```

6. Generar la llave para el cliente

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
Type the word 'yes' to continue, or any other input to abort.
  Confirm request details: yes

Using configuration from /home/jeferson/pki/openssl-easyrsa.cnf
Enter pass phrase for /home/jeferson/pki/private/ca.key:
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'cliente01'
Certificate is to be certified until Jul 26 17:36:09 2028 GMT (825 days)

Write out database with 1 new entries
Database updated

Notice
------
Certificate created at:
* /home/jeferson/pki/issued/cliente01.crt

Notice
------
Inline file created:
* /home/jeferson/pki/inline/cliente01.inline

jeferson@jeferson-IdeaPad-3-15ITL6:~$
```

Certificado auto-firmado para Apache

1. Instalar Apache

```
jeferson@jeferson-IdeaPad-3-15ITL6: ~
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/
systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service
 → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
jeferson@jeferson-IdeaPad-3-15ITL6:~$
```

2. Instalar el certificado para que lo use Apache
   a. Vincular los archivos de Easy-RSA con Apache

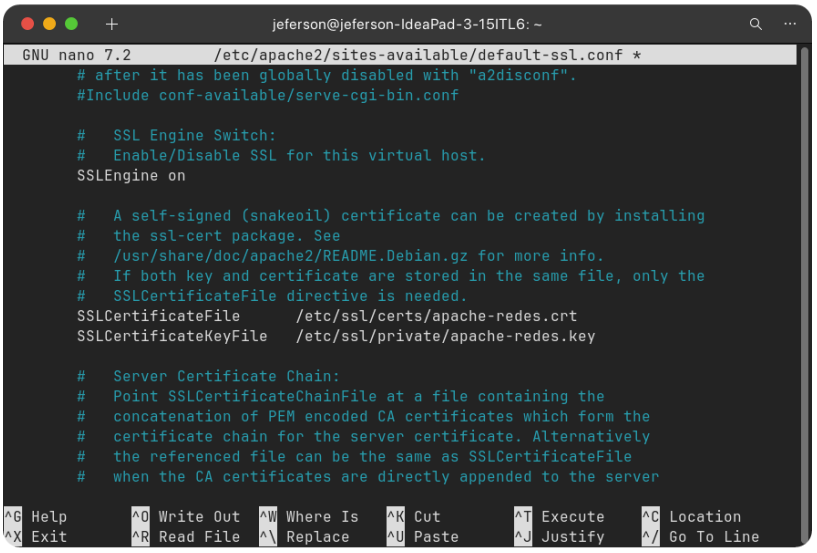
```
# Copiar el certificado del servidor (el que llamaste 'redes')
sudo cp ~/pki/issued/redes.crt /etc/ssl/certs/apache-redes.crt

# Copiar la llave privada del servidor
sudo cp ~/pki/private/redes.key /etc/ssl/private/apache-redes.key

# Copiar el certificado de la CA (necesario para que sea una cadena completa)
sudo cp ~/pki/ca.crt /etc/ssl/certs/ca-redes.crt
```

   b. activar el módulo de seguridad

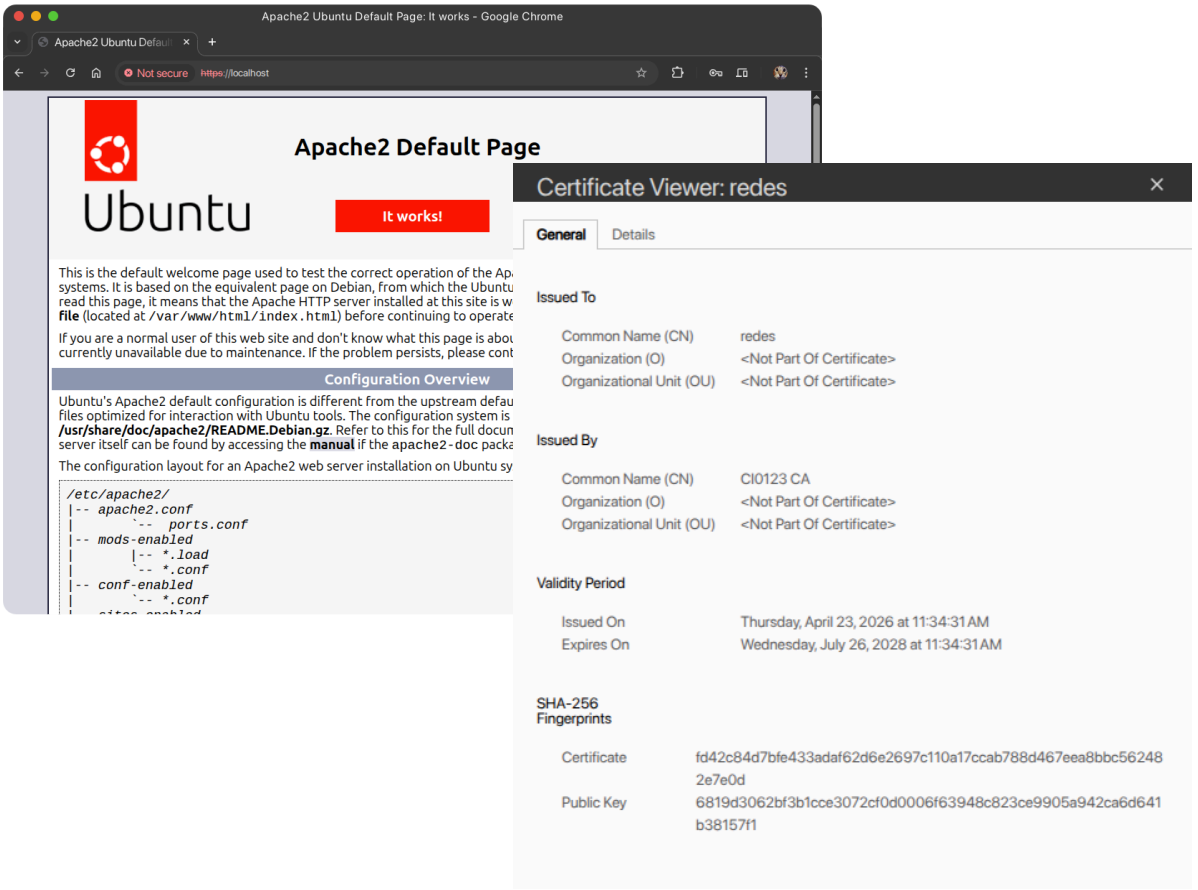
```
sudo a2enmod ssl
```

   c. Editar el archivo de configuración del sitio seguro:

```
SSLCertificateFile /etc/ssl/certs/apache-redes.crt
SSLCertificateKeyFile /etc/ssl/private/apache-redes.key
SSLCACertificateFile /etc/ssl/certs/ca-redes.crt
```



d. activar y reiniciar

```
sudo a2ensite default-ssl.conf
sudo systemctl restart apache2
```

3. Prueba de que el Apache está usando el certificado generado